Effect of Drip Irrigation and Soil Amendments on Water Use Efficiency in Cotton under Semi-Arid Conditions

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Abstract

This study evaluated drip irrigation and organic manure application on cotton grown during kharif season in Maharashtra under black cotton soil conditions.

Introduction

Water scarcity and soil degradation are major constraints for cotton production in semi-arid regions of India.

Results

Drip irrigation increased water-use efficiency by 24% compared with flood irrigation in cotton under semi-arid conditions.

Application of organic manure increased soil organic carbon by 0.3 g/kg relative to the control treatment over one cropping season.

Contour bunding reduced runoff and soil loss by 18% compared with the conventional tillage plots at the watershed outlet.

Soil pH increased from 6.2 to 6.8 following gypsum application on sodic soils.

Conclusion

Drip irrigation is recommended for cotton cultivation to improve water use efficiency under semi-arid conditions. Farmers should adopt contour bunding to reduce soil loss on sloped watershed lands.